

Video Transcript

Video 8: Time Loops (03:20)

When you do simulation, some sort of animation, say a visualization or in our case automation, you want to move through some time interval, and that is precisely what we're going to be doing. As you can see here, I am defining a template for a time loop. You can see there that I have a function, a number of tasks, and at the end of it I have a function that is part of the language that allows you to create a timer. It says the time at which you will call the function that you provide, and then you can fire it. As you can see there, that function is calling itself. So, we will have a loop. So, let's go ahead and write up that code. Let's go ahead and open up the timer file. We'll start off by importing time and timer. After that, we will create or we will define our tasks. These are dummy tasks, as you can see there. I'm simply printing to the console to the terminal task 1, and I'm going to make it go to sleep for a couple of seconds simply to simulate something that is taking some time to complete. Then I'm going to go ahead and add one more, which will be task 2. And then I'm going to have my function that is like the template we just discussed that has a number of tasks, and then that timer that allows us to call it ourselves. So, we inside of that function, we will execute task one or perform task one, then two.

Then we will write a timestamp to the terminal. And then we will set our timer to execute in five seconds, call itself and start. So, let's go ahead and run this. So, let's go ahead and run that file... Python 3... timer. And as you can see, nothing happened there. Let me check and see that I saved. Yes, and I can see the problem. I did not invoke the time loop, so it runs through everything, but it doesn't do anything. So, let's go ahead and save, run once again. As you can see there, immediately we start to execute task one, then task 2. Notice that there is a delay performing each one of them. Then we get our timestamp and then we start over task 1, task 2. Then we execute our next timer that calls itself and note there the time difference. As you can see there we have 54 seconds, then we are at zero-four seconds, then at 14 seconds. And the reason for that is that if you add up the seconds between the pauses there, you can see there that we have five here on line 16, then we have three on line 10 and then two in line six. So, all of them added up together are 10 seconds. So, if we look at the time stamps there, they are 10 seconds apart. So, as you can see, this this will run forever. It is a quick and dirty way to do a

time loop, and it'll work fine for our objective, which is to automate a number of tasks.

